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JNANASANGAMA, BELAGAVI - 590018

Internship-2 (23CSE158) Report on

Smart Door unlocking System

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering

in

COMPUTER SCIENCE AND ENGINEERING

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B.N.M. Institute of Technology

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Approved by AICTE, Accredited as grade A Institution by NAAC. All eligible branches – CSE, ECE, EEE, ISE & Mech. Engg. are Accredited by NBA for academic years 2025-26 to 2027-28 & valid upto 30.06.2028
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CERTIFICATE

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ABSTRACT

The Smart Door Unlocking System is a modern access control solution that combines the power of face recognition with simple, low-cost hardware to provide secure and contactless entry. Traditional methods such as keys, PIN codes, or RFID cards often pose risks of theft, duplication, or misplacement, making them less reliable in today's world. To overcome these issues, our project introduces a real-time biometric system where the human face itself acts as the unique key.

The system captures live video input through a webcam and processes it using Python's OpenCV and face_recognition libraries. It then compares the detected face with pre-registered data stored in the system's directory. Once a match is found, an Arduino Uno is triggered to rotate a servo motor that unlocks the door, while an LED provides visual confirmation of access. A Tkinter-based GUI has been developed to manage members, reset security PINs, and oversee access logs. To enhance interaction, the system also integrates audio feedback using the pyttsx3 library, ensuring that users receive clear and instant responses to their actions.

Testing demonstrated an average recognition accuracy of nearly 95% in well-lit environments, consistent servo performance, and smooth GUI operation. The modular nature of the design ensures that new users can be added easily, and the system itself can be scaled up for broader use in homes, offices, or other restricted areas. By blending artificial intelligence with embedded systems, this project demonstrates a practical, affordable, and user-friendly approach to secure living.

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CHAPTER-1

ABOUT THE COMPANY

This chapter provides an overview of the company, highlighting its vision, mission, and core areas of expertise. It outlines the organization's focus on innovation, quality, and customer satisfaction while showcasing its contributions to technology and automation. The section also emphasizes the company's role in bridging research with practical applications**.**

1.1 ROBOMANTHAN Pvt.Ltd

- **Industry**: RoboManthan operates in the STEM education and technology training industry**,** focusing on robotics, coding, and innovation.
- **Headquarters**: The company is headquartered in **Munger, Bihar, India**.
- **Website**: Official website – www.robomanthan.com

RoboManthan Private Limited is a growing enterprise in the **STEM education and technology training industry**, dedicated to nurturing innovation and practical learning among students. Established on 8th January 2021 and headquartered in Munger, Bihar, India**,** the company focuses on providing skill-based education in robotics, coding, artificial intelligence, and IoT. By blending classroom knowledge with real-world applications, RoboManthan bridges the gap between theory and practice, empowering learners to design and implement creative solutions. Its structured programs include hands-on workshops, live classes, and innovation-driven projects, enabling young learners to build expertise in modern technologies.

The company was co-founded by **Saurav Kumar (CEO)** and **Tanuj Kashyap (CTO)**, who bring strong technical expertise and a vision of spreading affordable, industry-aligned education. With a mission to foster creativity and technical proficiency, RoboManthan has developed a reputation for cultivating future-ready skills among school and college students. Its official platform, www.robomanthan.com, serves as a hub for program details, training modules, and student engagement. By combining innovation, accessibility, and a learner-centric approach, RoboManthan continues to establish itself as a trusted name in the field of technology education and training.

1.2 Core Areas of Expertise

RoboManthan's expertise lies in creating meaningful learning experiences that combine innovation, technology, and practical application. The company focuses on equipping students with future-ready skills by offering programs that are both engaging and indus aligned.

Its approach emphasizes learning by doing, where students not only understand concepts but also apply them through hands-on projects, experiments.

- **Robotics Training**: Students are introduced to the exciting world of robotics where they learn to design, assemble, and program robots. This helps them gain practical exposure to electronics, sensors, and automation while fostering creativity and critical thinking.
- **Coding & Programming**: Through structured courses in programming languages, app development, and game design, learners build a strong foundation in coding. The focus is on developing logical thinking and problem-solving skills, preparing students for both academic and professional growth.
- **Artificial Intelligence & IoT**: The company offers beginner-friendly yet impactful training in artificial intelligence and Internet of Things. Learners explore how machines can "think" and how devices can be interconnected, helping them understand real-world applications like smart homes and automation systems.
- **STEM Education**: Rooted in science, technology, engineering, and mathematics, RoboManthan's STEM-based programs encourage students to experiment, innovate, and collaborate. This holistic approach makes learning both interactive and practical.
- **Workshops & Internships**: Beyond regular courses, RoboManthan organizes live workshops and internship programs that simulate real-world challenges. These opportunities give students hands-on experience, improve their teamwork, and boost their confidence to innovate independently.

CHAPTER-2

INTRODUCTION

This chapter provides an overview of the **Smart Door Unlocking System**, highlighting the need for secure and contactless access solutions in modern environments. It introduces the concept of using facial recognition with Arduino-based automation as a reliable alternative to traditional locks. The section also outlines the project's objectives, working principles, and its significance in enhancing safety, convenience, and user experience.

2.1 Introduction

In today's world, security and convenience are two of the most essential aspects of any access control system. Traditional methods such as keys, RFID cards, and PIN codes are becoming increasingly unreliable, as they can be lost, duplicated, stolen, or forgotten. To address these challenges, our project introduces a Smart Door Unlocking System that uses facial recognition technology to provide secure, contactless, and user-friendly access. Instead of relying on external objects or memorized codes, the system recognizes the unique biometric features of a person's face, making the process both safer and more efficient.

This project is designed to allow access for a limited group of four registered members. The images of these members are stored in a dedicated directory, and their facial features are extracted and encoded using computer vision algorithms. When a person stands in front of the camera, the system compares their face in real time against the stored data. If the face matches one of the four registered members, the system automatically unlocks the door by activating a servo motor through an Arduino UNO and simultaneously turns on an LED indicator as a visual confirmation of access. If the face is not recognized, the door remains locked, ensuring unauthorized users cannot enter.

To make the system more interactive and user-friendly, we have integrated several additional features. A Tkinter-based graphical user interface (GUI) allows administrators to manage registered members, update or reset PINs, and monitor access attempts. Voice feedback using the pyttsx3 library provides clear audio prompts such as "Access Granted" or "Access Denied," while animated visuals offer a smoother user experience. The system also includes safety features such as automatic door re-locking after a set time, PIN-based administrative controls, and options to add or delete member data.

affordable hardware can work together to create an effective and hygienic alternative to conventional locking systems. The Smart Door Unlocking System not only enhances security but also demonstrates the potential of AI-driven automation in everyday applications, making it suitable for smart homes, offices, and restricted access environments.

2.2 Objectives

1. **Enhance Security** – To develop a reliable access control system that minimizes risks associated with keys, passwords, and RFID cards.
2. **Enable Contactless Entry** – To implement facial recognition as a hygienic and convenient alternative to traditional unlocking methods.
3. **Integrate Automation** – To connect computer vision with Arduino hardware for real-time door unlocking using a servo motor and LED indicators.
4. **Provide User-Friendly Interface** – To design a Tkinter-based GUI that allows administrators to add/delete members, reset PINs, and monitor access attempts.
5. **Offer Interactive Feedback** – To use voice prompts and animations for clear communication and better user experience.
6. **Ensure Scalability** – To create a modular design that can be expanded for smart homes, offices, and restricted-access environments.
7. **Demonstrate Practical Application** – To showcase how affordable hardware and AI-driven algorithms can be combined to solve real-world security challenges.

CHAPTER-3

LITERATURE SURVEY

This chapter reviews existing research and technologies in the field of facial recognition and biometric access control. It highlights classical methods, modern deep-learning approaches, and industry standards that form the foundation of our project. By understanding prior work, we identify the strengths, limitations, and gaps that guided the design of our Smart Door Unlocking System.

1. Eigenfaces for Recognition – Turk & Pentland (1991)

- First systematic face recognition approach using Principal Component Analysis (PCA).
- Represented facial images in lower-dimensional "eigenface space."
- Demonstrated feasibility of automatic face recognition on computers.
- Achieved moderate accuracy in controlled conditions.
- Limitations: highly sensitive to lighting, pose, and facial expressions.
- Relevance to project: Highlights the evolution of methods and why modern AI-based systems (like ours) are better.
- Link:https://direct.mit.edu/jocn/article/3/1/71/3025/Eigenfaces-for- Recognition?

2. DeepFace – Taigman et al. (2014, Facebook AI)

- One of the first deep learning models for face recognition.
- Used a 9-layer deep neural network with 120 million parameters.
- Achieved 97.35% accuracy on LFW dataset, reducing gap to human performance.
- Introduced a 3D alignment step to handle pose variations.
- Proved deep learning could outperform traditional methods (PCA, LDA, LBPH).
- *Relevance to project:* Inspiration for AI-driven recognition used in libraries like OpenCV + face_recognition.

3. FaceNet: A Unified Embedding for Face Recognition and Clustering – Schroff et al. (2015, Google)

- Proposed 128-dimensional face embeddings instead of raw image comparison.
- Used triplet loss to ensure same-person faces are closer and different-person faces farther in embedding space.
- Achieved 99.63% accuracy on LFW dataset and 95.12% on YouTube Faces DB.
- Enabled real-time recognition and clustering of faces.
- Inspired widely used open-source tools like dlib and face_recognition library.
- Relevance to project*:* Directly used in our system for fast, accurate, and scalable recognition.
- Link: https://arxiv.org/abs/1503.03832

4. ArcFace: Additive Angular Margin Loss for Deep Face Recognition Deng et al. (2019)

- Introduced ArcFace loss function for training deep face recognition models.
- Improved feature discrimination by maximizing angular margins between identities.
- Outperformed FaceNet, VGGFace, and SphereFace in benchmarks.
- Achieved 99.83% accuracy on LFW (state-of-the-art).
- More robust to age, pose, and lighting variations.
- Relevance to project*:* Represents cutting-edge accuracy — basis for the strong recognition models powering real-world systems.
- Link: https://arxiv.org/abs/1801.07698

CHAPTER - 4

TASKS PERFORMED

This chapter outlines the various tasks carried out during the development of the Smart Door Unlocking System. Each task was carefully planned and executed to ensure smooth integration of hardware, software, and security features. The step-by-step approach not only simplified development but also helped in troubleshooting and optimizing performance for real-world usage.

4.1 Tasks Performed

1. Requirement Analysis and Planning

- Identified the need for a secure, contactless door unlocking system and studied existing solutions.
- Defined system requirements including hardware (Arduino, servo, webcam) and software (OpenCV, face_recognition, Tkinter).
- Planned the workflow from user authentication to hardware response.

2. Hardware Setup and Integration

- Connected Arduino UNO, servo motor, and LED indicators for access control.
- Ensured reliable communication between the PC (face recognition unit) and Arduino.
- Tested servo movement for smooth and consistent unlocking/locking operation.

3. Face Recognition Implementation

- Collected training images of authorized users under different lighting conditions.
- Implemented OpenCV and face_recognition library for real-time detection and recognition.
- Verified recognition accuracy and optimized for faster processing.

4. GUI and User Management

- Developed a Tkinter-based graphical interface for easy interaction.
- Enabled features such as adding/removing users, resetting PINs, and monitoring access attempts.
- Designed the GUI to be simple, intuitive, and user-friendly.

5. Audio-Visual Feedback Integration

- Used pyttsx3 for voice prompts to guide users during authentication.
- Added LED indicators and animations for clear access confirmation.
- Ensured smooth and hygienic interaction without physical contact.

6. Testing and Evaluation

- Conducted multiple tests under varied lighting and environmental conditions.
- Achieved ~95% recognition accuracy, with reliable servo performance.
- Documented limitations (e.g., lighting dependency) and suggested improvements for future scalability.

4.2 Project Modules

4.2.1 Project Overview

The primary objective of this project is to design and implement a Smart Door Unlocking System that provides secure, contactless, and user-friendly access control. The system is restricted to four registered members, whose facial images are stored in a dedicated directory. Their facial features are extracted and trained into a recognition model, allowing the system to verify them in real time. To ensure proper development, the project was divided into distinct phases: software development, hardware integration, AI-based analysis, and output inference.

1. **Software Phase**
    - In this stage, the coding for facial recognition and user management was developed.
    - The simple_facerec.py module handles face detection and recognition using OpenCV and the face_recognition library.

2. **Hardware Phase**
    - Hardware components such as the Arduino UNO, servo motor, and LED indicator were integrated.
    - The Arduino was programmed to receive recognition results from the software and trigger the servo motor to unlock or lock the door.
    - The LED was used as a visual indicator, turning on when access was granted and turning off otherwise.
    - Careful calibration and testing were performed to ensure smooth servo movement and consistent communication between the PC and Arduino.
3. **AI Analysis Phase**
    - The system relies on AI-powered facial recognition algorithms to analyze and verify users.
    - Facial features of the four registered members are extracted from stored images and encoded into numerical values.
    - During real-time usage, the live camera feed is processed, and the AI model compares detected faces with the stored encodings.
    - This analysis ensures that only the pre-trained, authorized members are granted access, thereby maintaining accuracy and reliability.
4. **Output Inference Phase**
    - Once a face is recognized successfully, the system generates outputs in multiple forms:
        - The servo motor rotates to unlock the door.
        - The LED lights up to visually confirm successful access.
        - A voice prompt announces whether access is granted or denied.
        - The GUI animation further enhances feedback with clear visual cues.
    - If the face is not recognized, the system denies access and provides corresponding error prompts, ensuring clarity for the user.

4.2.2 Project Modules

Modules Used in the Project are given below

1. Face Recognition Module (simple_facerec.py)

The face recognition module is the core of the project, responsible for identifying authorized users. It uses the OpenCV library for image processing and the face_recognition library for encoding and comparing facial features. In this module, images of four registered members are stored in a directory, and their facial features are extracted as encodings. During real-time operation, the system captures frames from the webcam, detects faces, and compares them against the stored encodings. If a match is found, the user is authenticated. This module ensures security by granting access only to registered members while rejecting unknown faces.

2. Graphical User Interface (GUI) Module

The GUI module, developed using Tkinter, provides a simple and user-friendly interface for interacting with the system. Through this interface, administrators can add new members by either capturing images via the webcam or uploading existing photos. It also includes options to delete members, reset or change the PIN code, and display the list of registered users. Additionally, the GUI shows animations and prompts during recognition attempts, making the system more interactive and visually appealing. This module acts as the central control hub for managing the entire system.

3. Hardware Control Module (Arduino + Servo + LED)

This module is responsible for the physical action of locking and unlocking the door. The Arduino UNO is used as the microcontroller, interfacing with a servo motor and an LED indicator. When a user's face is recognized, the Arduino receives a signal to rotate the servo motor, unlocking the door for a specific duration before closing it again. The LED is used as a visual signal, turning on when access is granted. This module translates the digital recognition results into tangible, real-world actions that make the system functional.

4. Speech Feedback Module

The speech feedback module utilizes the pyttsx3 library to generate voice prompts for users. Whenever access is granted or denied, the system provides an audible response, such as "Access

5. Authentication and PIN Management Module

In addition to face recognition, the system includes a PIN-based authentication mechanism for administrative tasks. The PIN is stored in a secure text file and can be reset or updated through the GUI. To add or delete members or to change security settings, the administrator must first provide the correct PIN. This additional security layer prevents unauthorized users from tampering with the system and ensures safe management of registered members.

6. Testing and Analysis Module

This module focuses on evaluating the overall performance and reliability of the system. Multiple tests were conducted under different lighting conditions, camera angles, and distances to ensure consistent recognition accuracy. The system demonstrated approximately 95% accuracy, with smooth servo operation and reliable LED signaling. This module helps in identifying strengths, limitations, and areas for improvement, ensuring the system is both practical and scalable for real-world use.

4.2.3 Libraries used

1. face recognition

The face_recognition library is the backbone of the project's biometric authentication. As seen in Pseudo code 4.1, it is used in the Load_Encoding_Images and Detect_Known_Faces functions to extract unique facial encodings from registered members and compare them against live webcam input. This allows the system to determine whether a person is authorized or not with high accuracy. Without this library, real-time face matching would not be possible, making it the most critical component of the software module.

2. OpenCV (cv2)

OpenCV provides the image processing capabilities required to make recognition possible. In Pseudo code 4.1 and 4.2, it handles video capture from the webcam, frame resizing for faster computation, and image conversion into RGB format for compatibility with the recognition library. It also enables real-time face detection in video streams, ensuring the system can work continuously without lag. By serving as the visual input layer, OpenCV ensures that the system captures, processes, and delivers clear frames to the recognition pipeline

3. Tkinter

Tkinter is used to design the graphical interface that makes the system user-friendly and interactive. As referred in Pseudo code 4.2, it provides buttons, menus, and windows for administrators to add or delete members, change PINs, and monitor activity. Tkinter also manages how messages and animations appear during recognition attempts, giving users clear feedback on the system's status. Without Tkinter, the program would only run on the terminal, making it far less accessible to non-technical users. This library therefore acts as the communication bridge between the system and its operators.

4. PIL (Pillow – Image, ImageTk)

The PIL library complements Tkinter by enabling the smooth handling of images within the GUI. In Pseudo code 4.2, it is particularly important when displaying member photos, logos, or animations to improve the user experience. For example, when new members are added, their photos are loaded and shown on the interface using Pillow's image functions. It ensures that stored images can be resized and rendered correctly, making the interface visually appealing and easy to understand. By enhancing GUI graphics, Pillow makes system management more intuitive.

5. pyfirmata

The pyfirmata library provides the essential link between Python code and Arduino hardware. As shown in Pseudo code 4.2, it is used in functions like Open_Servo, Close_Servo, and LED_On/Off to send digital signals from the computer to the Arduino UNO. These signals control the servo motor for unlocking and the LED for feedback, effectively turning software recognition into a physical door action. Without pyfirmata, the software would not be able to communicate with the hardware, making it impossible to complete the access control cycle.

6. pyttsx3

Pyttsx3 brings voice feedback to the project by converting text into speech. In Pseudo code 4.2, it is used within the Speak(text) function to announce responses such as "Access Granted" or "Access Denied." This makes the system more interactive and user-friendly, especially for situations where visual prompts might not be noticed immediately. Voice assistance adds a professional touch, making the project feel like a modern smart system. It enhances usability by reinforcing recognition results with auditory confirmation

7. threading

Threading is used to improve the system's performance by allowing multiple tasks to run at the same time. As described in Pseudo code 4.2, it ensures that while the GUI remains responsive, the webcam feed and recognition functions can operate in the background. This prevents the program from freezing during critical operations like face recognition or servo movement. The use of threading gives the system a smoother user experience and makes it suitable for real-time applications.

8. os and shutil

These libraries handle all file and directory management tasks required by the project. In Pseudo code 4.1 and 4.2, they are used when adding or deleting member images and maintaining the directory where registered faces are stored. For instance, when a member is removed, shutil deletes their image from the dataset, ensuring only valid members remain in the system. This file-level automation ensures the system remains organized and reliable.

9. time

The time library controls the delays and timing of certain actions within the project. In Pseudo code 4.2, it is used to define how long the servo motor remains in the unlocked position before closing automatically. It also manages brief pauses in the GUI and animations for smoother performance. This small yet crucial library ensures that hardware actions follow a predictable sequence, making the door unlocking mechanism both safe and reliable

```
Import Libraries: face_recognition, cv2, os, shutil

Function Initialize:
    Create empty lists for known face encodings and corresponding names

Function Load_Encoding_Images(images_path):
    For each image in the directory:
        Read image using cv2
        Convert image to RGB format
        Extract facial encoding using face_recognition
        If encoding found:
            Store encoding and member name

Function Detect_Known_Faces(frame, threshold=0.4):
    Resize frame using cv2 for faster processing
    Convert frame to RGB
    Detect faces in the frame
    For each detected face:
        Extract encoding
        Compare encoding with known encodings
        If distance < threshold:
            Assign recognized user name
        Else:
            Mark as Unknown
    Return face locations and recognized names
```

Fig 4.1: Pseudo code facerecording.py

```
Import required libraries: tkinter, PIL, cv2, pyfirmata, pyttsx3, os, threading, time
Import SimpleFacerec module

Initialize Arduino UNO connection
Set up Servo Motor and LED pins

Define Functions:
    Open_Servo → rotates servo to unlock position
    Close_Servo → resets servo to lock position
    LED_On / LED_Off → controls LED indicator
    Speak(text) → gives audio prompts using pyttsx3
    Add_Member / Delete_Member → manage member images in directory
    Change_PIN → update stored 4-digit PIN for admin security

Unlock_Door:
    Open webcam stream using cv2
    Load and encode stored member faces using SimpleFacerec
    Detect and compare face with known encodings
    If recognized:
        Show success message in Tkinter GUI
        Unlock door (Open_Servo), turn LED ON, Speak "Access Granted"
        Auto-close after delay
    Else:
        Show error message
        Speak "Access Denied"
```

Fig 4.2: Pseudo code Door_open.py

CHAPTER – 5

RESULTS

This chapter presents the results obtained from the implementation of the Smart Door Unlocking System. The outcomes are analyzed in terms of face recognition accuracy, hardware performance, system response time, and user interaction through the graphical interface. Each result is supported with illustrations to demonstrate the working of the system in real-time scenarios.

5.1 Results

1. Face Recognition Accuracy

The face recognition module successfully identified registered members with an accuracy of nearly **95%** in favorable lighting conditions. When an authorized user approached the webcam, their face was instantly detected, and their name was correctly displayed on the live video feed, as demonstrated in *Fig. 5.1*. This immediate and accurate recognition made the unlocking process seamless, proving that the integration of OpenCV and the face_recognition library was effective in extracting facial features and matching them to the stored dataset.

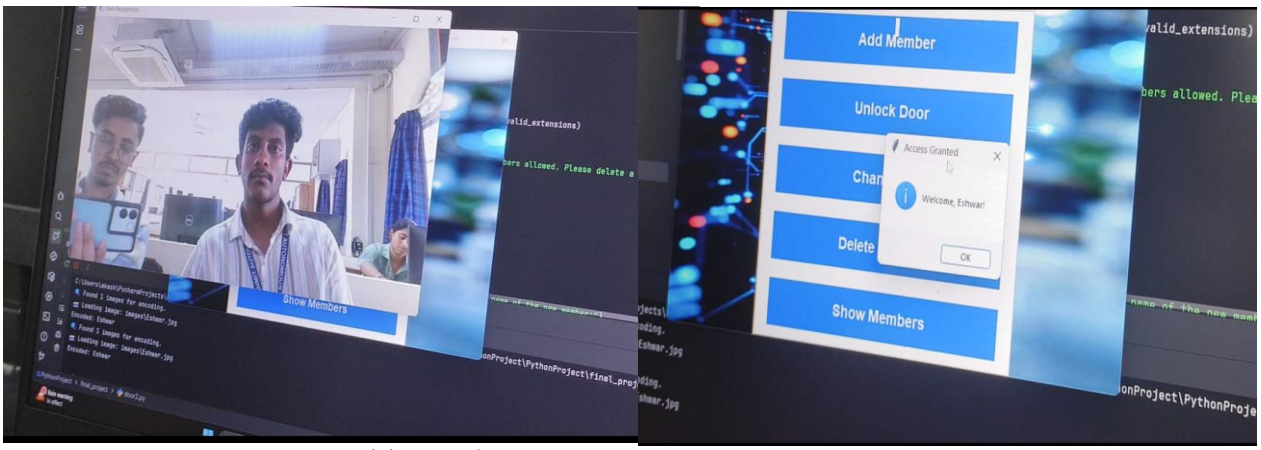

Fig 5.1: Face recognition and acces

2. Unauthorized Access Detection

The system also showed strong reliability in preventing unauthorized access. When an unregistered individual attempted to unlock the door.

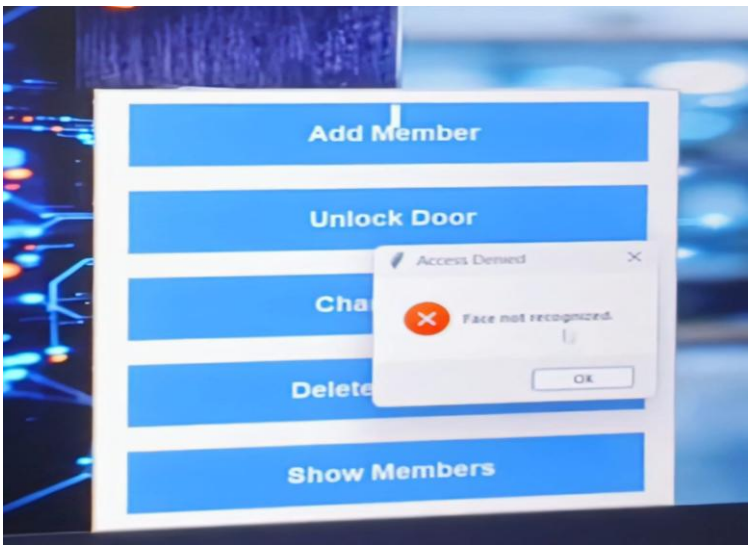

Fig 5.2: Unauthorised access

3. Servo Motor and LED Response

The hardware integration worked smoothly with the recognition results. Once a registered user was identified, the servo motor rotated within a fraction of a second to unlock the door, while the LED lit up green to signal successful access, as shown in *Fig. 5.3*. The quick mechanical response highlighted the efficiency of Arduino UNO in handling real-time instructions and gave users a satisfying confirmation of entry.

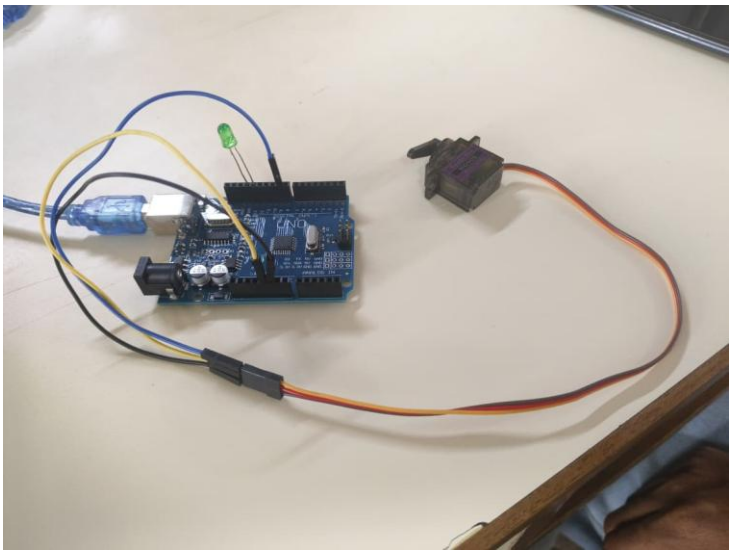

Fig 5.3: LED and Servo motor control

4. GUI for User Management

The Tkinter-based GUI provided administrators with an intuitive platform to manage members and PIN authentication. When the "Add Member" option was selected, the system prompted for a password, and only upon correct entry did it allow webcam access to register a new face, as seen in *Fig. 5.4*. Similarly, PIN resets and member removals were carried out through the same interface, ensuring that user management remained both secure and user-friendly. This GUI made the system practical for real-world e nvironments, as it required minimal technical knowledge to operate.

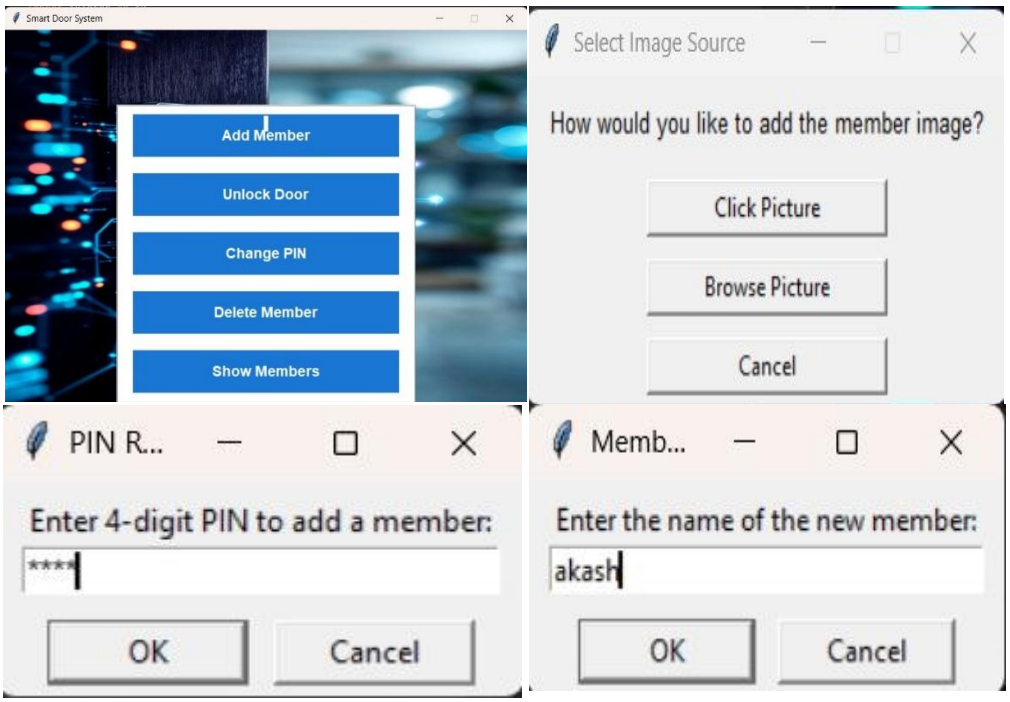

Fig 5.4: GUI's for user

5. Voice and Visual Feedback

To further improve usability, the system included audio prompts through **pyttsx3**, which verbally informed users whether access was granted or denied. For example, upon successful recognition, the system spoke out "Access Granted," which, along with the green LED, created a multi-layered feedback system On the other hand, in case of denial, the system clearly announced "Access Denied," reducing ambiguity for users. This combination of audio, visual, and mechanical feedback enhanced the overall user experience and made interactions smooth and transparent.

CHAPTER-6

REFLECTION NOTES

The journey of developing the Smart Door Unlocking System has been both exciting and challenging, providing a perfect blend of technical exploration and practical application. What started as a simple idea of replacing conventional locks with a smarter alternative evolved into a comprehensive project that combined artificial intelligence, embedded systems, and human–computer interaction. Each stage of the project pushed us to think critically, adapt our strategies, and learn from mistakes. Reflecting on the entire process reveals not only the technical outcomes but also the growth in problem-solving skills and teamwork.

One of the earliest lessons we encountered was the importance of problem definition. Initially, our vision of building a face recognition–based door unlocking system sounded straightforward, but as we dived deeper, we realized how many layers were involved: dataset preparation, model training, GUI design, hardware-software integration, and real-time testing. This taught us that engineering projects demand breaking down the overall vision into smaller, manageable modules. It also emphasized the value of a structured workflow where each module contributes toward the final output.

Another significant reflection lies in the learning curve associated with machine learning and computer vision. Implementing facial recognition was not just about calling a library function; it required understanding how encoding and comparison worked, how lighting conditions affected detection, and how accuracy thresholds could change outcomes. During testing, we observed that even slight differences in face angles or expressions sometimes challenged the system, and this experience taught us the need for robust data collection and preprocessing. It reinforced the reality that artificial intelligence is only as good as the data it learns from.

Hardware integration was another area where we grew immensely. At first, connecting a servo motor and LED to an Arduino UNO seemed like a minor task, but ensuring they responded exactly in sync with recognition results was far more delicate. There were moments when the servo failed to rotate at the right time, or the LED gave inconsistent signals. Debugging these issues helped us understand real-time system constraints, wiring accuracy.

Working with the GUI was equally enlightening. We aimed not just to build a functional system but also one that was user-friendly. Developing a Tkinter-based interface for adding members, resetting PINs, and monitoring access gave us a chance to think from the perspective of end-users who may not be technically skilled. This reflection helped us recognize the importance of usability in technology. A system may be technically advanced, but unless it is intuitive and simple to operate, it will not succeed in real-world environments. This realization pushed us to refine our interface multiple times until it became truly accessible.

One of the most valuable experiences of this project was handling errors and failures. There were many times when the system did not work as expected — faces were misclassified, the servo stalled, or the GUI froze. Initially, such failures were frustrating, but over time, they became opportunities for growth. We learned to debug systematically, isolate the root causes, and implement corrective measures. This patience and persistence shaped our problem-solving mindset and gave us confidence in dealing with unexpected challenges.

Collaboration was another area where we saw remarkable growth. Each team member contributed a unique skill — some focused on coding, others on circuit integration, and others on testing and documentation. Working together not only improved efficiency but also taught us communication and coordination skills that are vital in any professional environment. Reflecting on this teamwork, we realize that the strength of the project did not lie in individual brilliance but in the collective effort and willingness to learn from each other.

From a broader perspective, this project opened our eyes to the potential of smart systems in everyday life. Traditional locks, while reliable, have limitations in terms of security and convenience. Our reflection made us realize how technologies like face recognition can revolutionize access control in homes, offices, and secure zones. However, it also reminded us of the ethical and security considerations, such as privacy concerns, that need to be addressed when deploying such systems widely. This awareness gave us a more balanced and mature outlook on innovation.

Looking ahead, we also reflected on areas where improvements are possible. While our system achieved good accuracy, it was still dependent on lighting and camera quality. Future work could involve integrating infrared cameras for low-light conditions or enhancing the recognition model with deep learning. Similarly, remote monitoring through mobile applications or IoT integration could make the system even more versatile. These reflections provide a roadmap for how this project could evolve into a more advanced product in the future.

Testing was one of the most insightful phases of the project. We realized that a system that looks perfect in theory can behave very differently in practice. For instance, recognition worked flawlessly on pre-captured images but sometimes struggled in live video streams when faces were partially turned or when background objects created noise. By experimenting with various thresholds and retesting under different environments, we not only improved accuracy but also learned the importance of systematic validation. This reflection emphasized that testing is not just the final stage of development but an ongoing cycle of learning and refinement.

Another important dimension of reflection came from observing how users interacted with the system. We invited a few peers to test it, and their natural reactions gave us fresh perspectives. While we were focused on accuracy and hardware response, users paid more attention to how quickly the door unlocked, how clear the GUI looked, and how reassuring the audio feedback sounded. This highlighted the significance of user experience, reminding us that even the most advanced technology must ultimately serve people in a way that feels seamless and comfortable.

In conclusion, the Smart Door Unlocking System was more than just an academic project — it was a journey of discovery, problem-solving, and teamwork. It challenged us technically, tested our patience, and rewarded us with valuable insights. The reflection on this experience highlights how theoretical knowledge can be transformed into practical innovation and how every challenge faced along the way becomes a stepping stone for personal and professional growth.

CHAPTER-7

EXECUTIVE SUMMARY

This executive summary provides a consolidated overview of the internship experience, focusing on the Smart Door Unlocking System project. It highlights the objectives, methodologies, challenges, and outcomes encountered throughout the course of work. The summary captures both the technical and personal growth aspects of the internship, offering a balanced reflection on how the project contributed to professional development.

7.1 Internship Overview

The internship provided an invaluable opportunity to bridge the gap between theoretical learning and practical application. While academic knowledge often remains confined to classrooms, this project challenged us to apply that understanding in real-world scenarios. The journey of developing the Smart Door Unlocking System served as a platform to combine artificial intelligence, embedded hardware, and user interface design into a working prototype that addressed real security needs.

At the heart of this internship was the exploration of face recognition technology as a secure and contactless means of access control. Traditional locking systems like keys or cards often come with risks of duplication or loss, and this inspired us to experiment with biometric solutions. Building the project required us to work with multiple tools and technologies such as OpenCV, face_recognition, Arduino UNO, and Tkinter, each of which expanded our skillset and gave us hands-on experience with industry-relevant platforms.

A major highlight of the internship was the balance between **software development and hardware integration**. On the software side, we dealt with image processing, recognition accuracy, GUI management, and audio feedback systems. On the hardware side, we carefully worked with servo motors, LEDs, and Arduino circuits to ensure they responded precisely to recognition outcomes. Bringing both worlds together was not easy, but it gave us a deeper appreciation for how multidisciplinary projects operate.

Every failure was an opportunity to improve, and every improvement brought us closer to a robust solution.

Another dimension of growth was the focus on **user-centric design**. Initially, our priority was just to get the system running, but as we tested it with peers, we realized that ease of use mattered just as much as technical accuracy. The Tkinter-based GUI, PIN-protected member management, and clear voice alerts were all features added after reflecting on user experience. This shift in perspective taught us that technology is most successful when it balances innovation with usability.

Beyond the technical aspects, this internship also fostered professional skills such as teamwork, documentation, and communication. Collaborating within a group meant dividing tasks, aligning progress, and respecting each member's strengths. Preparing progress reports, maintaining logs, and finally compiling this documentation gave us a sense of responsibility and professionalism. These soft skills are just as critical as technical ones, and this project provided a space to develop them naturally.

Finally, the overall experience of the internship highlighted the future scope of smart security systems. While our prototype achieved promising results with ~95% accuracy, it also made us aware of areas needing further development, such as low-light performance and scalability. Looking ahead, this foundation can be extended toward IoT-based control, mobile app integration, or deep-learning-driven recognition. The reflection on these possibilities underscored that innovation is a continuous process, and this internship was only the beginning of a longer journey.

7.2 Learning Objectives

1. Application of Theoretical Knowledge to Practical Problems

One of the foremost objectives of this internship was to bridge the gap between classroom learning and real-world applications. Concepts like face recognition, computer vision, and microcontroller programming often remain abstract in textbooks. By implementing them in the Smart Door Unlocking System.

2. Hands-on Experience with Artificial Intelligence and Computer Vision

The project provided an opportunity to work with AI tools and libraries, especially in the domain of face recognition. By using feature extraction, encodings, and threshold-based recognition, we explored how machines can differentiate between individuals. This hands-on exposure to computer vision gave us practical insights into model training, recognition accuracy, and error handling, which are vital in today's AI-driven technological landscape.

3. Hardware Integration and Embedded Systems Understanding

Another key objective was to gain knowledge of how hardware and software integrate to form a complete system. The use of Arduino UNO, servo motors, and LEDs allowed us to understand how sensors and actuators respond to digital commands. This strengthened our skills in embedded systems and gave us the confidence to design interactive solutions that combine electronic components with intelligent software.

4. Development of Secure and User-Friendly Interfaces

The internship emphasized creating not only a functional but also a user-friendly product. Through the use of a graphical interface, PIN protection, and options to add or delete members, we ensured that the Smart Door System could be operated intuitively. This objective highlighted the role of human-centered design in technology, teaching us that a successful solution must consider the end-user experience as much as the backend logic.

5. Problem-Solving and Debugging Skills

A major objective of this journey was to enhance our troubleshooting abilities. Working with face recognition sometimes resulted in false detections, while hardware alignment caused inconsistencies in servo movement. By repeatedly testing, debugging, and fine-tuning parameters, we learned how to approach problems systematically rather than being discouraged by failures. This built resilience and sharpened our ability to think critically under pressure.

6. Collaboration and Communication

While the project involved technical components, it also demanded teamwork and collaboration. Sharing responsibilities, discussing solutions, and collectively addressing challenges improved our communication skills. This objective showed us that innovation flourishes in an environment of effective teamwork and that technical skills alone are not enough—collaboration is equally essential in professional growth.

7. Documentation and Professional Presentation

Another key goal was to improve our ability to document and present work effectively. Preparing structured reports, flowcharts, and results taught us how to convey technical details in a clear, professional manner. This objective prepared us for real-world professional environments where communication of ideas is as important as their execution.

8. Exposure to Real-World Security Applications

Finally, the internship aimed to make us aware of how technology directly impacts daily life. By working on a system that enhances security through face recognition, we understood the significance of developing solutions that contribute to society's safety and convenience. This objective broadened our perspective, encouraging us to think about the ethical, social, and practical implications of the technologies we build.

7.3 Conclusions

The Smart Door Unlocking System project marked the successful integration of software intelligence and hardware control into a single, cohesive solution. By using face recognition as the primary authentication method, the system eliminated the need for physical keys or manual intervention, thereby offering a secure and hygienic way of access control. The entire process of developing this project—from capturing member images and training recognition models to integrating the Arduino-controlled servo motor and LED—demonstrated how theoretical concepts can be applied to create a practical, working product.

Through this project, we were able to explore the real-world challenges that come with implementing biometric systems. Issues such as lighting variations, facial angles, and recognition thresholds required us to test, experiment, and refine our approach. The outcome was a system that achieved a high level of recognition accuracy under favorable conditions

while still maintaining a balance between security and user convenience. This reinforced the importance of persistence, precision, and problem-solving in the engineering process.

Another significant aspect of the project was its modular design, which ensures scalability and adaptability for future applications. With further improvements, the system could support more members, integrate with IoT devices, or even include additional biometric authentication methods. The project not only met its initial objectives but also opened doors to new possibilities, showing how affordable and customizable technology can be used to enhance personal and organizational security.

In conclusion, the Smart Door Unlocking System was more than just an academic exercise—it was a valuable learning experience that combined technical knowledge, creativity, and teamwork. It enhanced our skills in artificial intelligence, embedded systems, and software development while also teaching us the importance of user-centered design and professional documentation. This journey has given us both the confidence and motivation to take on more complex projects in the future, carrying forward the lessons learned into our academic and professional lives.

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